

SR Latch

level triggered.
latched the door.

$$\begin{aligned}\overline{0+0} &\rightarrow 1 \\ \overline{0+1} &\rightarrow 0 \\ \overline{1+0} &\rightarrow 0 \\ \overline{1+1} &\rightarrow 0\end{aligned}$$

present-state =
Next state =

i/p s		i/p	o/p
<u>S</u>	<u>R</u>	<u>Q_n</u>	<u>Q_{n+1}</u>
0	0		
0	1		
1	0		
1	1		

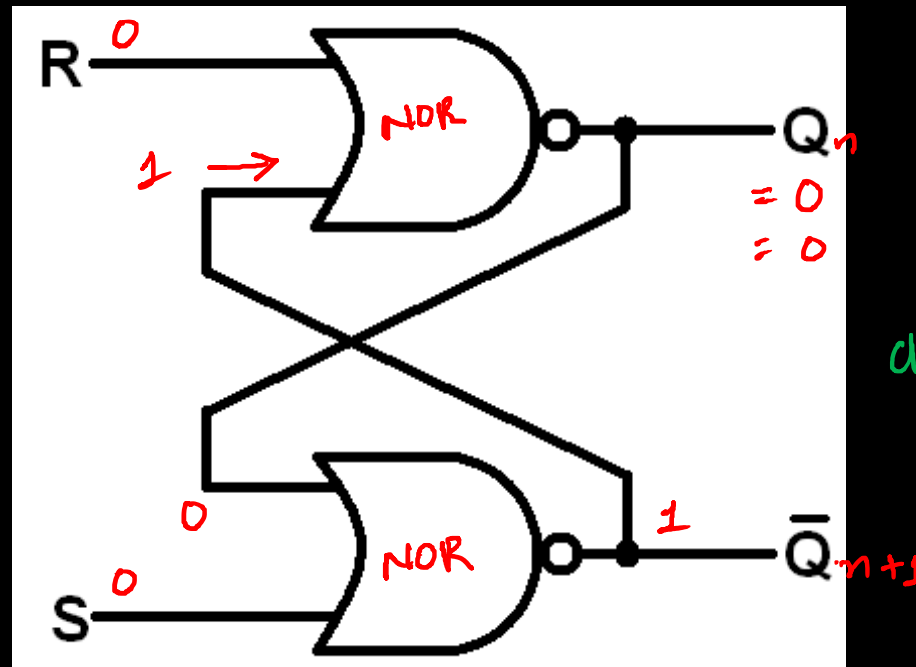
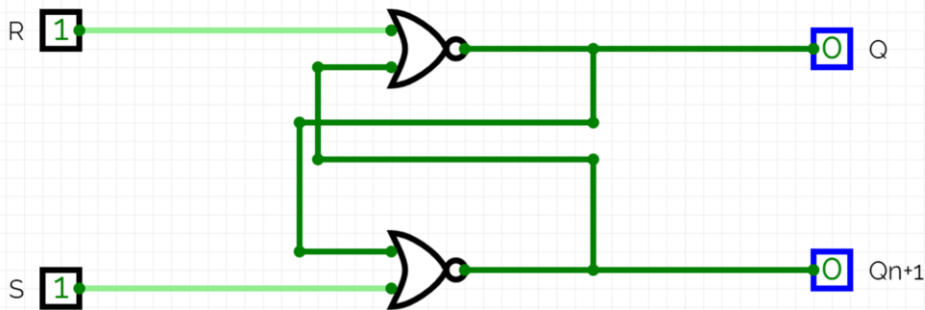
$$2^3 = 8 [0-7]$$

000
↓
111

Characteristic Table.

ANALYSIS.

Digital circuit



Latches <i>Easy to design</i> → <i>Level sensitive</i> → <i>Race condⁿ</i> <i>less P consumption</i> → <i>signal level.</i> <i>Asynch. Generally.</i>	Flip-Flops <i>consume more P</i> ← <i>Edge sensitive</i> → <i>Signal transitions</i> → <i>Predictable & Reliable.</i> <i>Synchrn. Generally</i>
The working manner of a latch is asynchronous which means, the output produced from the latch will depend on the input.	These are mainly used to store the binary data. An FF will have an extra CLK signal to make it work in a different way when contrasted with a latch. <i>+ All TIME.</i>
A latch doesn't contain any clock signal	A flip-flop contains a clock signal <i>+ Complex circuitry.</i>
Latches are very fast → ①	Flip-Flops (FFs) are very slow → ①
Latches consume less power → ② A Latch may be clockless or clocked → ③	FFs consume more power → ② A flip-flop can be clocked for all time → ③
The working of the Latches can be done by only binary inputs	FFs work by binary inputs & the CLK signal.
Latches are not capable of working as a register because of the lack of a CLK signal.	FFs are capable because they come by a CLK

Advantages of Latches

The **advantages of latches** include the following.

- The designing of latches is very flexible when we compare with **FFs (flip-flops)**
- The latches utilize less power.
- The performance of latch in the design of the high-speed circuit is quick because these are asynchronous within the design and there is no need of CLK signal.
- The shape of the latch is very small and occupies less area
- If the operation of latch based circuit is not finished in a set time, they borrow the necessary time from other to complete the operation

- Latches give aggressive clocking when contrasted with flip-flop circuits. → Remember this 'I' point: All others are same as previous FF.

Disadvantages of Latches

The **disadvantages of latches** include the following.

- There will be a chance of affecting the race condition, so these are less expected.
- When a latch is level sensitive, then there is a chance of meta-stability.
- Analyzing the circuit is difficult due to the property of level sensitive.
- The circuit can be tested by using an extra CAD program

A simple example of metastability can be found in an SR NOR latch, **when both Set and Reset inputs are true (R=1 and S=1) and then both transition to false (R=0 and S=0) at about the same time.** Both outputs Q and Q are initially held at 0 by the simultaneous Set and Reset inputs.

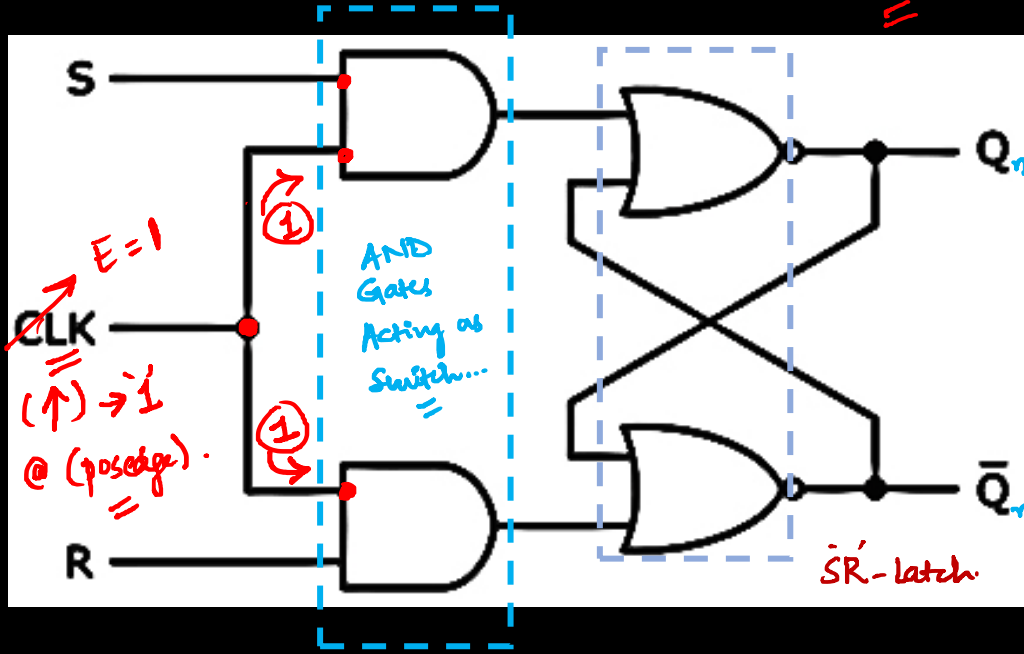
SR FlipFlop

SET ←

→ (S.T) =



0 → 0
1 → 1
0 → 1
1 → 0



E=1
(↑) → 1
@ (posedge).

AND Gates Acting as Switch...

CONTROL GATES.

RESET.

	Input		Output	
	S	R	Q_n	Q_{n+1}
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	-
7	1	1	1	-

MEMORY → SRAM (Read)

S	R	Q_n
0	0	Q_n
0	1	0
1	0	1
1	1	Invalid

CHARACTERISTIC TABLE

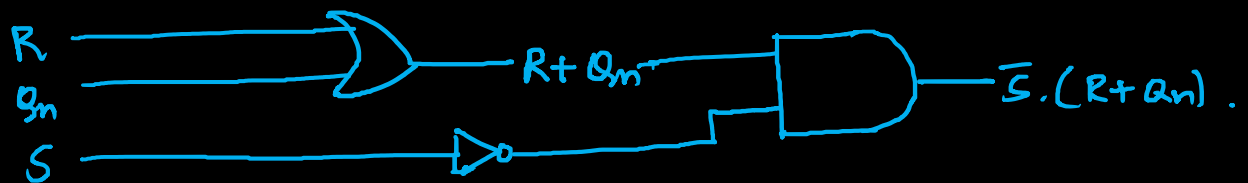
TRUTH-TABLE

'S': SET :- when $S = 1 \rightarrow Q_n \rightarrow$ set ; whatever value is there it's UNCHANGED.

'R': RESET :- when $R = 1 \rightarrow Q_n \rightarrow$ Reset : Q_n becomes zero ; As if Memory is clear.

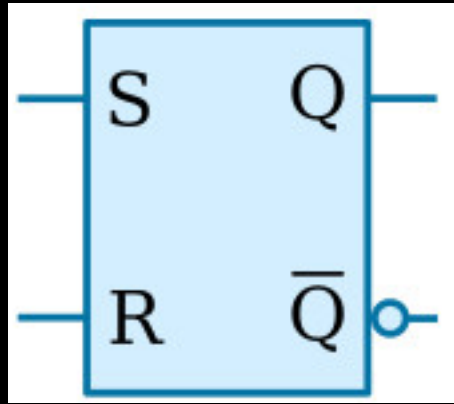
SR	00	01	11	10
Q_n				
0	0	1	3	2
1	1	1	7	6

$$Q_{n+1} = \bar{S}R + Q_n\bar{S} = \bar{S}(R + Q_n) : \text{CHARACTERISTIC Eqn.}$$



CHARACTERISTIC circuit

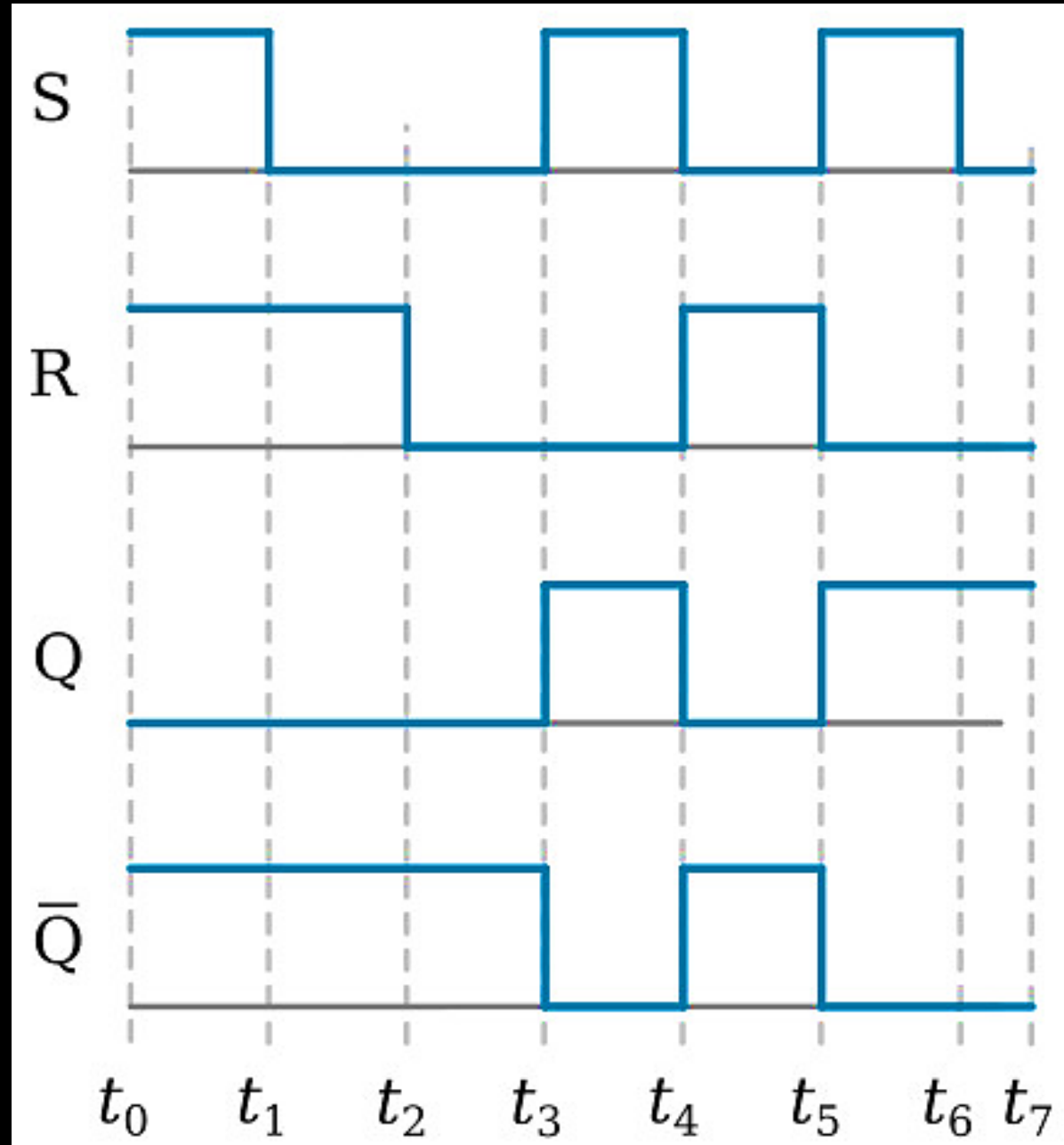
WAVEFORMS ANALYSIS



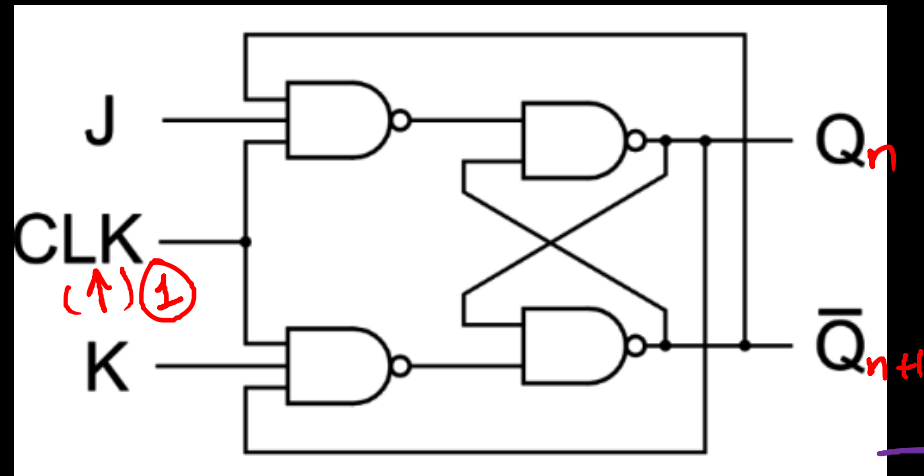
Characteristic table				
S	R	Q	$\bar{Q}$	Condition
0	0	*	*	latched
0	1	0	1	reset
1	0	1	0	set
1	1	X	X	jammed

* means previous state

X means 0 or 1



JK FlipFlop



Input			Output
J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

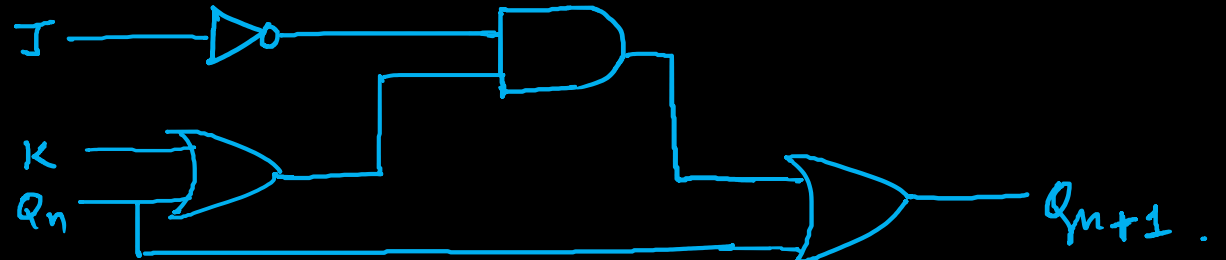
CHARACTERISTIC TABLE

J	K	Q_n
0	0	Q_n
0	1	0
1	0	1
1	1	$\bar{Q}_n \rightarrow$ Race Around Condition.

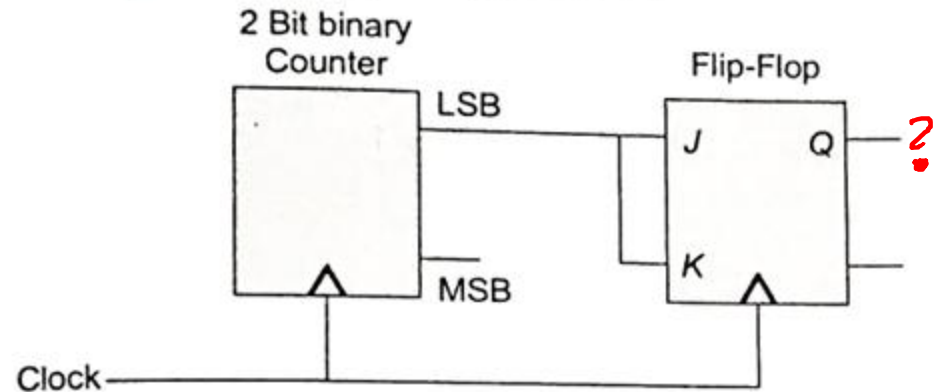
TRUTH-TABLE

J \ K	00	01	11	10
Q_n				
0	0	1	3	2
1	1	1	7	6

$$Q_{n+1} = \bar{J} \cdot K + \bar{J} \cdot Q_n + Q_n = \bar{J} \cdot (K + Q_n) + Q_n \text{ CHARACTERISTIC Eqn.}$$



5.55 For the circuit shown, the clock frequency is f_0 and the duty cycle is 25%. For the signal at the Q output of the Flip-Flop, .



- ✓ (a) frequency is $f_0/4$ and duty cycle is 50%
 (b) frequency is $f_0/4$ and duty cycle is 25%
 (c) frequency is $f_0/2$ and duty cycle is 50%
 (d) frequency is f_0 and duty cycle is 25%

[2022 : 2 Marks]

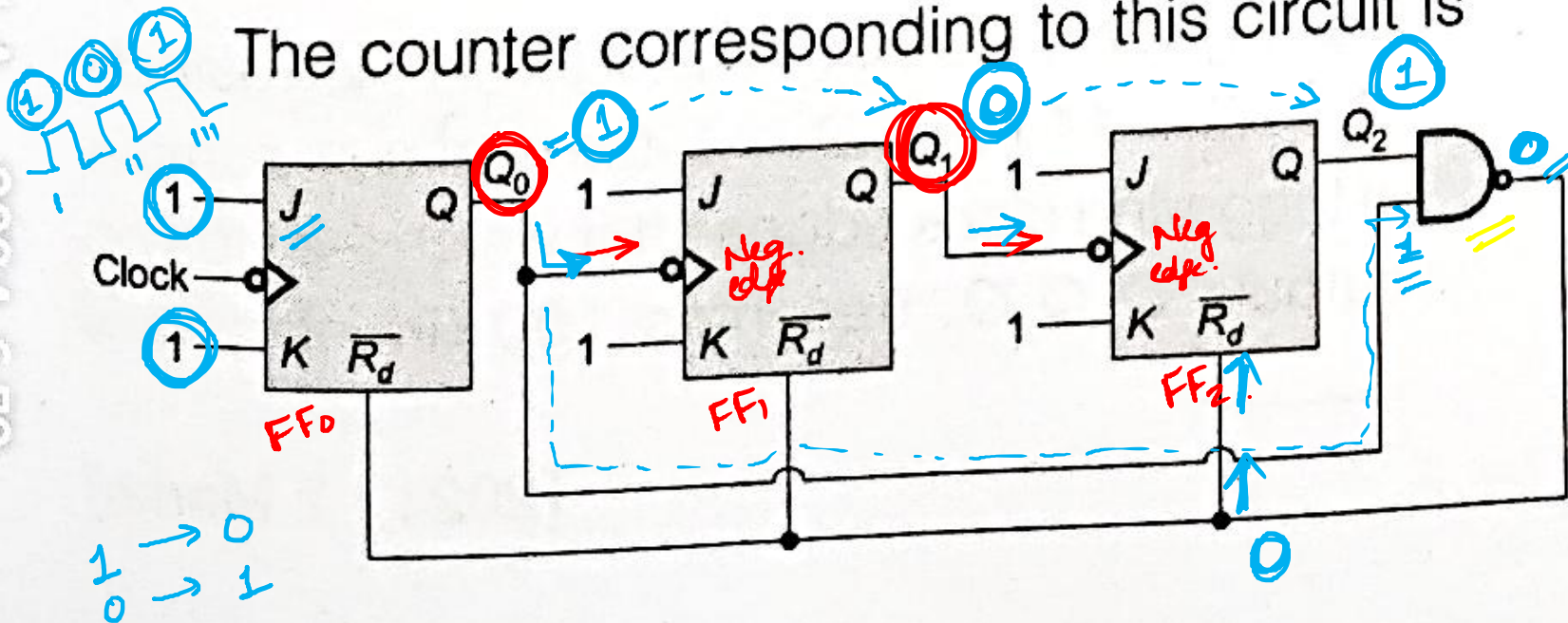


(i) No. of FF's = $\frac{1}{2} \times \frac{f_0}{2} \rightarrow \text{F.D.} = \frac{f_0}{4}$ ANS.

(ii) $25 \times 2 = \text{50 \%}$

5.45 The circuit shown consists of J-K flip-flops, each with an active low asynchronous reset ($\overline{R_d}$ input).

The counter corresponding to this circuit is



→ TOTAL NO. of FF's = 3.
that Means;

• Starting count = 000.
• Ending count = 111.

$$\left. \begin{array}{l} 2^3 - 1 \\ = 8 - 1 \\ = 7 \end{array} \right\}$$

	4	2	1
0 →	0	0	0
1 →	0	0	1
2 →	0	1	0
3 →	0	1	1
4 →	1	0	0
5 →	1	0	1
6 →	1	1	0
7 →	1	1	1

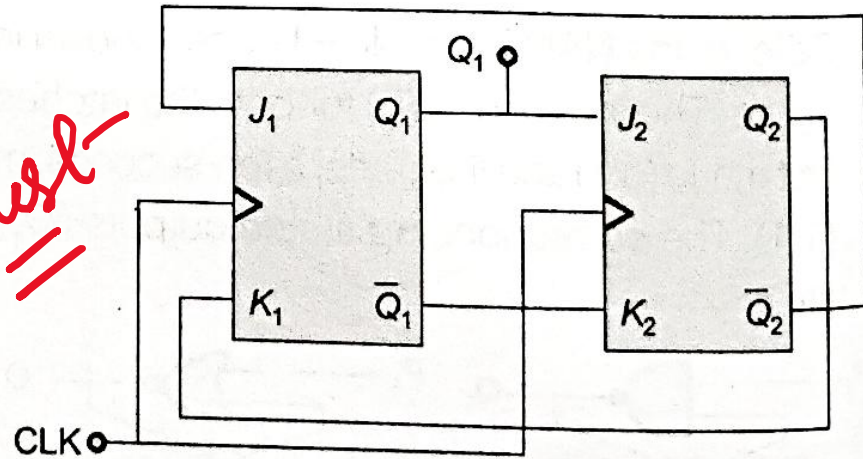
RESET

- (a) a modulo-5 binary up counter
 (b) a modulo-6 binary down counter
 (c) a modulo-5 binary down counter
 (d) a modulo-6 binary up counter

[2015 : 1 Mark, Set-3]

5.41 The outputs of the two flip-flops Q_1 , Q_2 in the figure shown are initialized to 0, 0. The sequence generated at Q_1 upon application of clock signal is

answer



(a) 01110...

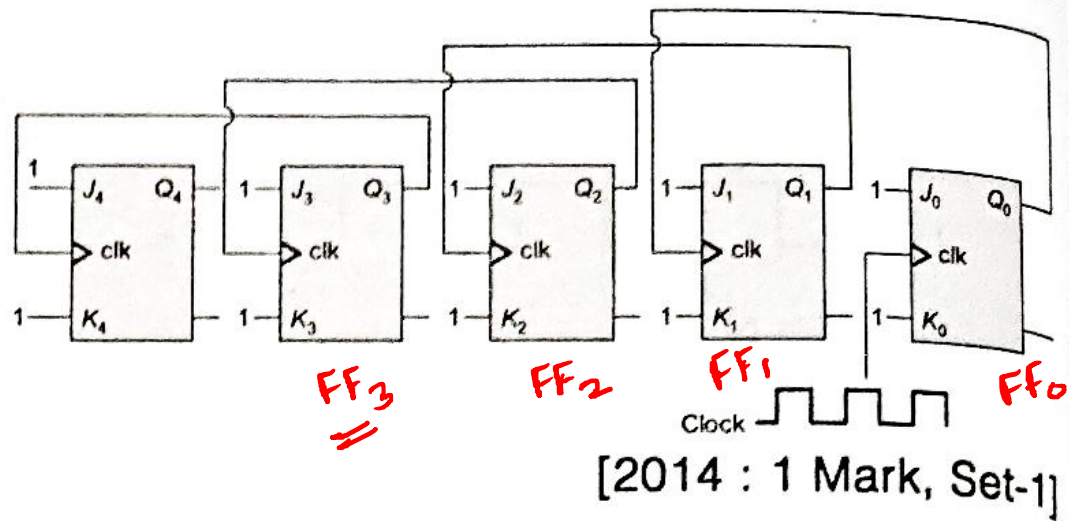
(b) 01010...

(c) 00110...

(d) 01100...

[2014 : 2 Marks, Set-2]

5.39 Five JK flip-flops are cascaded to form the circuit shown in Figure. Clock pulses at a frequency of 1 MHz are applied as shown. The frequency (in kHz) of the waveform at Q_3 is _____.



$$\odot \text{clk}(f) = 1 \text{ MHz}.$$

→ TOTAL NO. OF FFs = 4 → FD :

$$\frac{1 \text{ MHz}}{2^4} = \frac{1 \text{ MHz}}{16}$$

$$= 0.0625$$

$$= 62.5 \times 10^3 \text{ Hz}$$

$$= \underline{\underline{62.5 \text{ kHz}}}$$

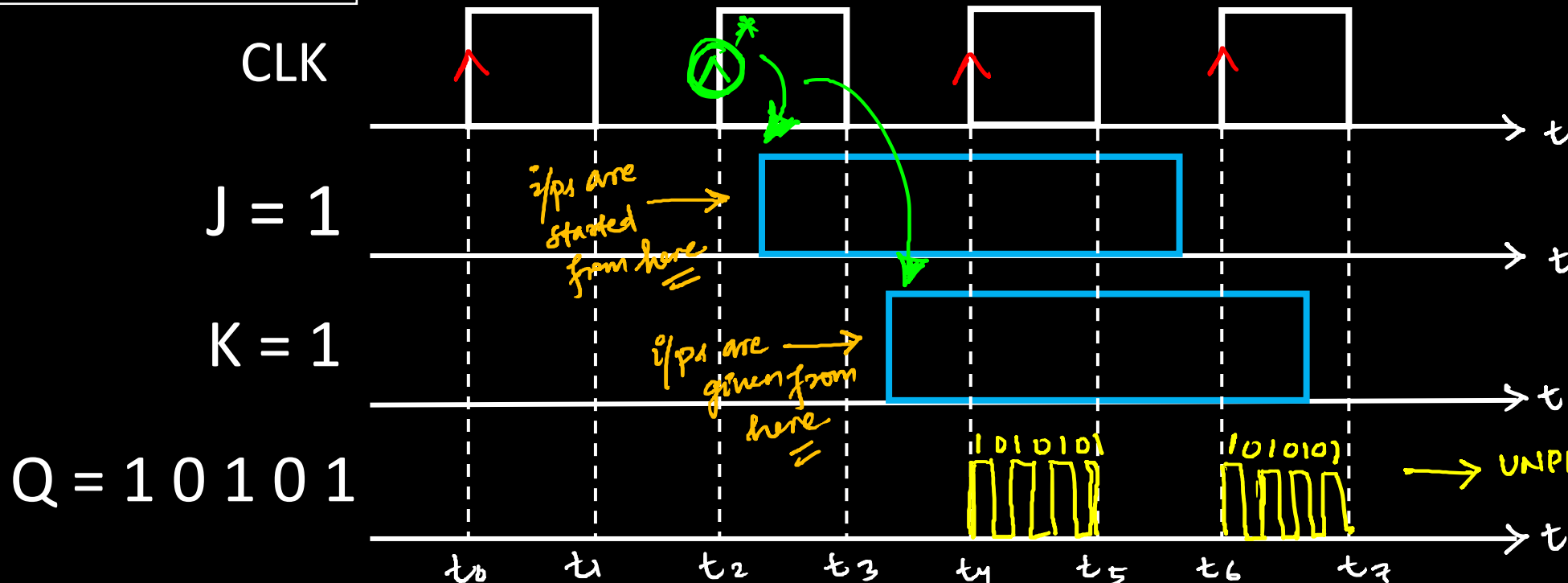
* At Every +ve Edge
of clock signal
and simultaneously
i/p's of 'J' & 'K'

$Q \rightarrow 10101 \checkmark$
 $\bar{Q} \rightarrow 01010 \checkmark$

The diagram illustrates the timing of a JK flip-flop. The inputs are J=1 and K=1. The clock (CLK) has a rising edge marked with a red arrow and labeled "(↑) +ve E.T". The output Q transitions from 1 to 0, and the complemented output $\overline{Q}$ transitions from 0 to 1. Red arrows indicate the setup and hold times relative to the clock edge.

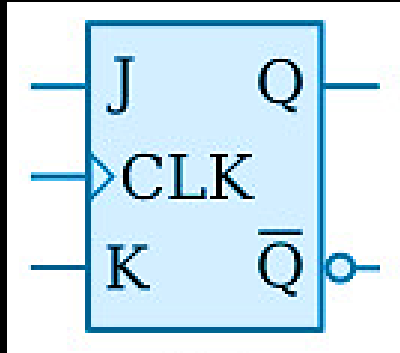
'is RACING'g

$0, 1, 0, 1, 0, \dots$



→ UNPREDICTABLE RANDOM
BOUNCING
→ t OUTPUT ...

WAVEFORMS ANALYSIS

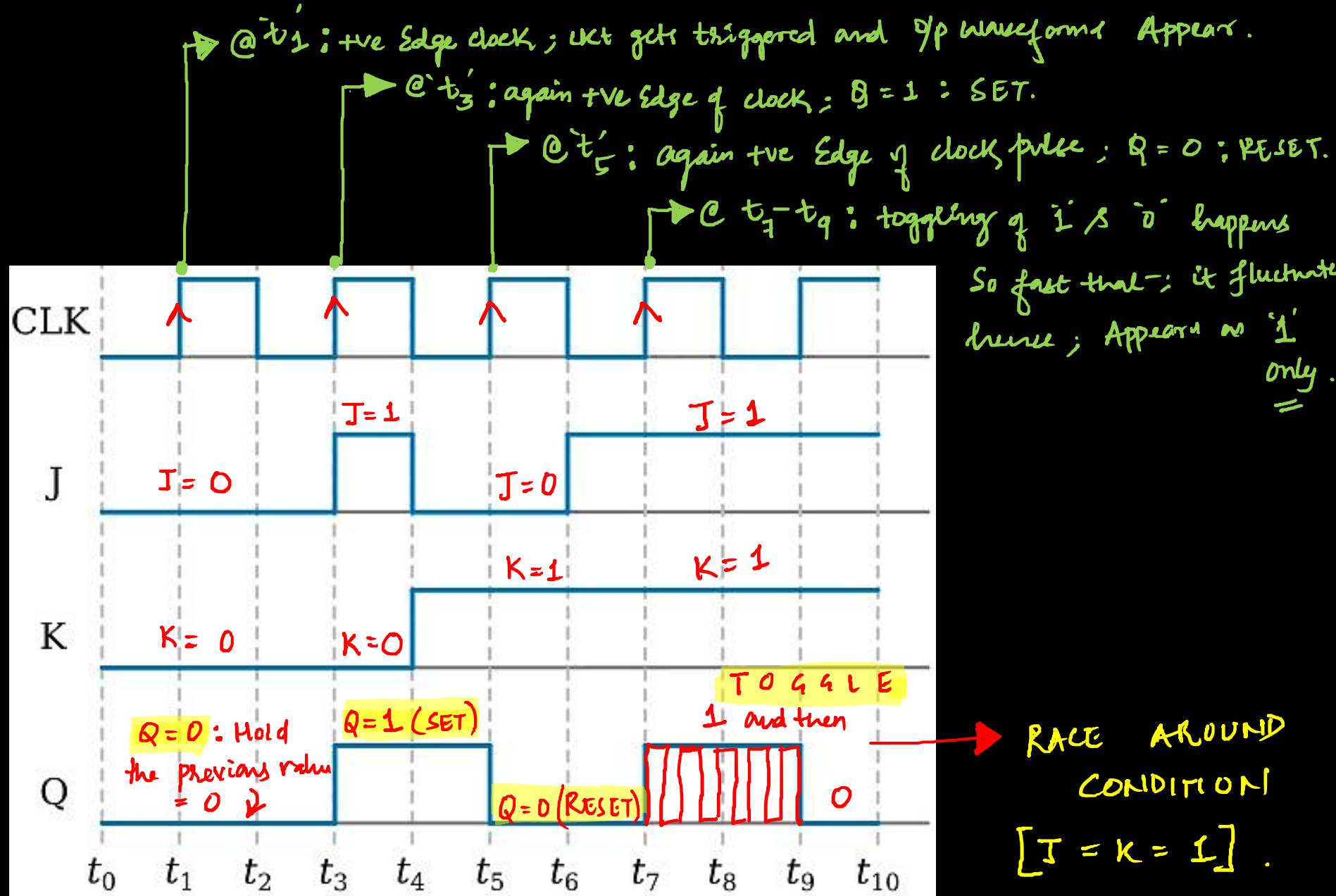


CLK	J	K	Q_{next}	Comment
Rising edge	0	0	Q	Hold state
Rising edge	0	1	0	Reset
Rising edge	1	0	1	Set
Rising edge	1	1	$\bar{Q}$	Toggle
Non-rising	X	X	Q	No change

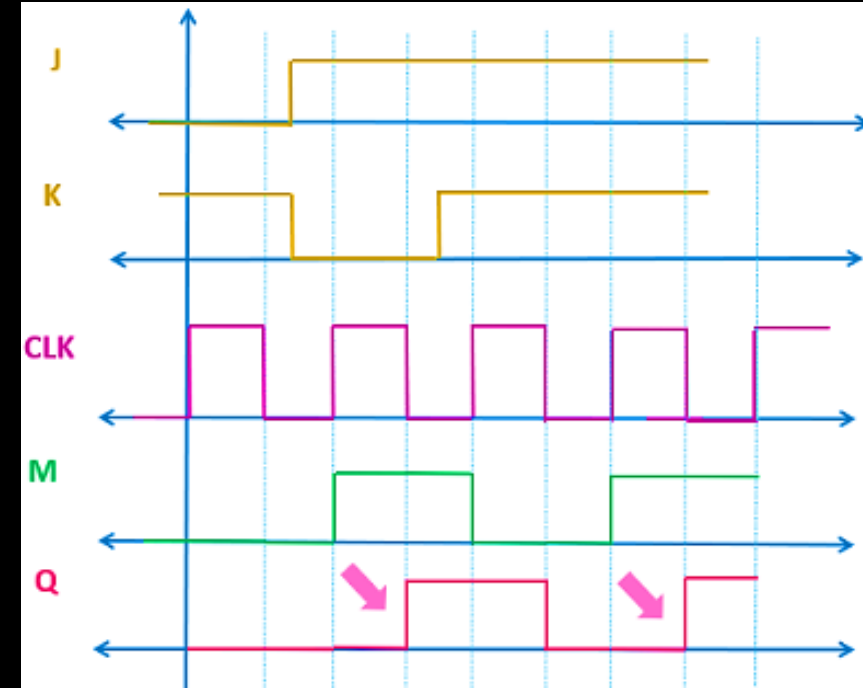
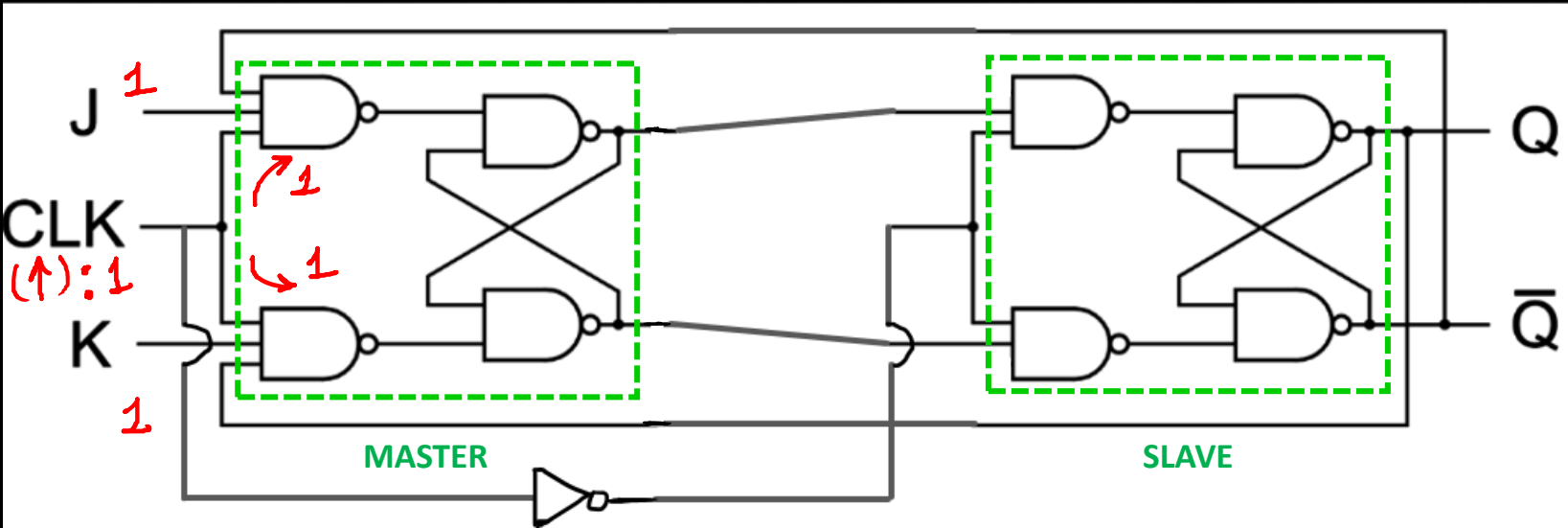
Q_{next} - "after the clock transition" output

Q - the current output

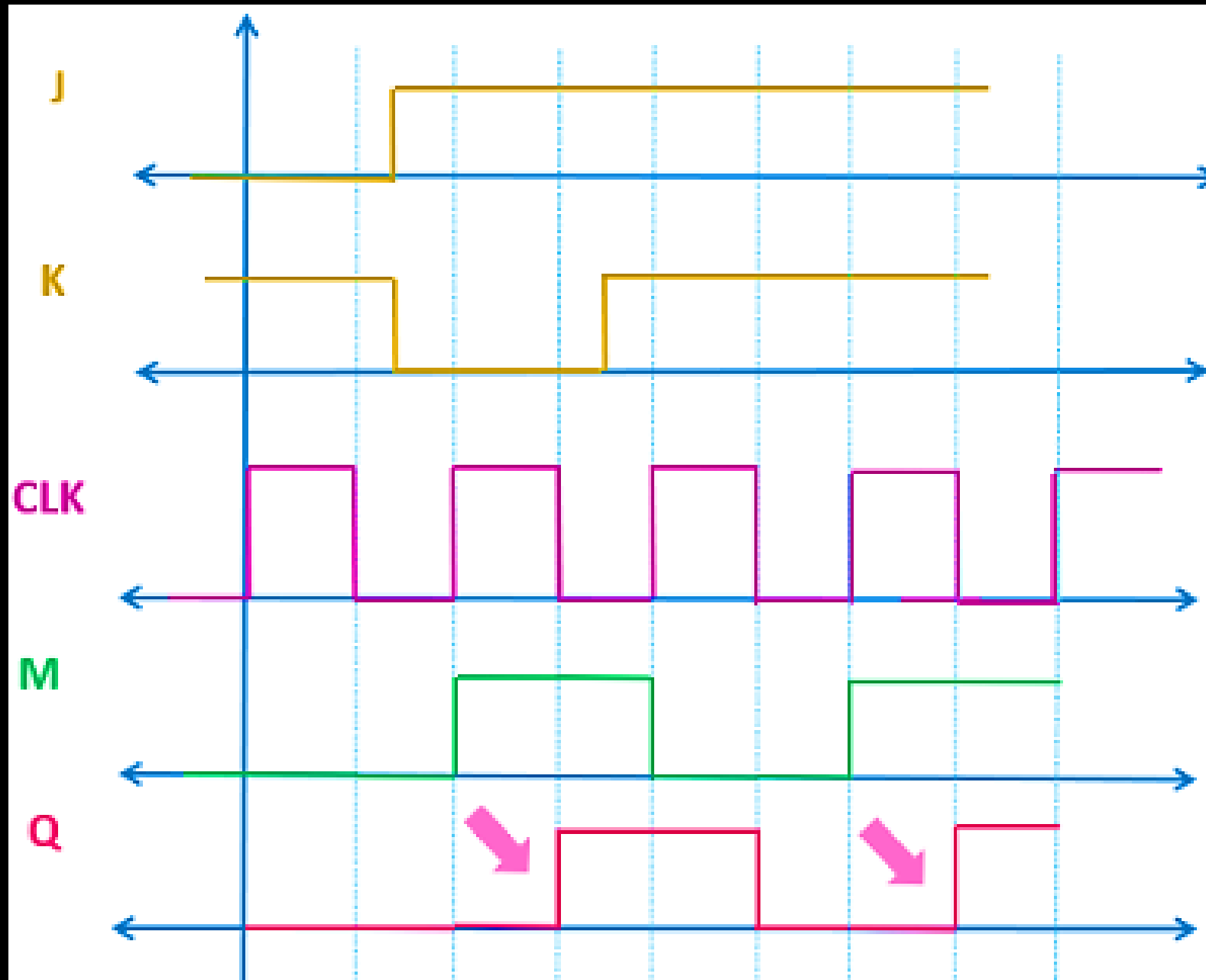
X - the signal is irrelevant



Master Slave JK FlipFlop



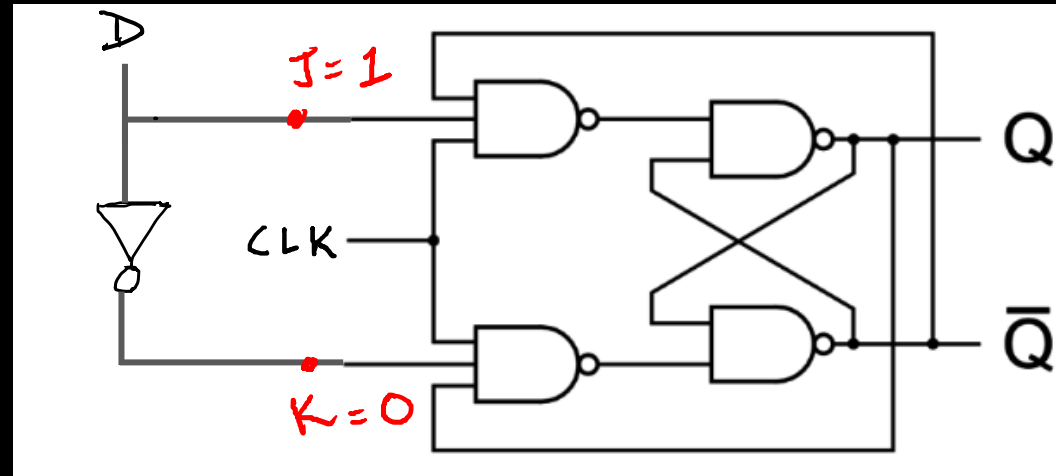
WAVEFORMS ANALYSIS



D FlipFlop

Input			Output
J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

$\left. \begin{matrix} 0 \\ 1 \\ 0 \end{matrix} \right\} Q_n$
 $\left. \begin{matrix} 1 \\ 1 \end{matrix} \right\} 1$
 $\left. \begin{matrix} 1 \\ 0 \end{matrix} \right\} \bar{Q}_n$



$\bar{D}$
 $\bar{D} :$

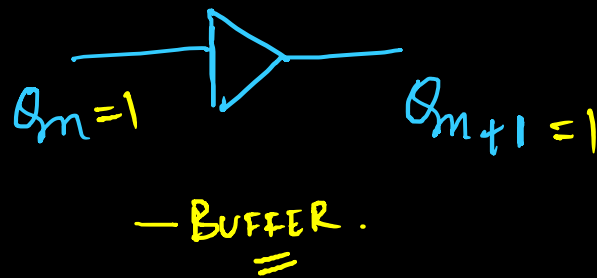
J	K	Q_n
0	0	Q_n
0	1	0
1	0	1
1	1	1

TRUTH-TABLE

CHARACTERISTIC TABLE

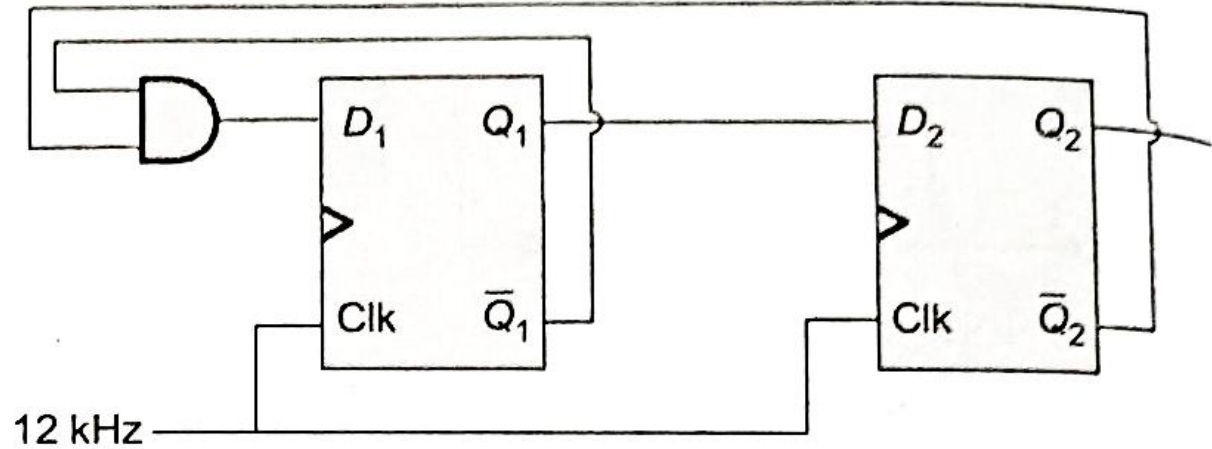
D	0	1
Q_n		
0	0	1
1	1	1

$Q_{n+1} = Q_n$ CHARACTERISTIC Eqn.



- DATA/DELAY Flipflop
 - SISO, SIPO, PISO, PIPO
 - Counters designing, FDS...

5.52 In the circuit shown, the clock frequency, i.e., the frequency of the Clk signal, is 12 kHz. The frequency of the signal at Q_2 is _____ kHz.



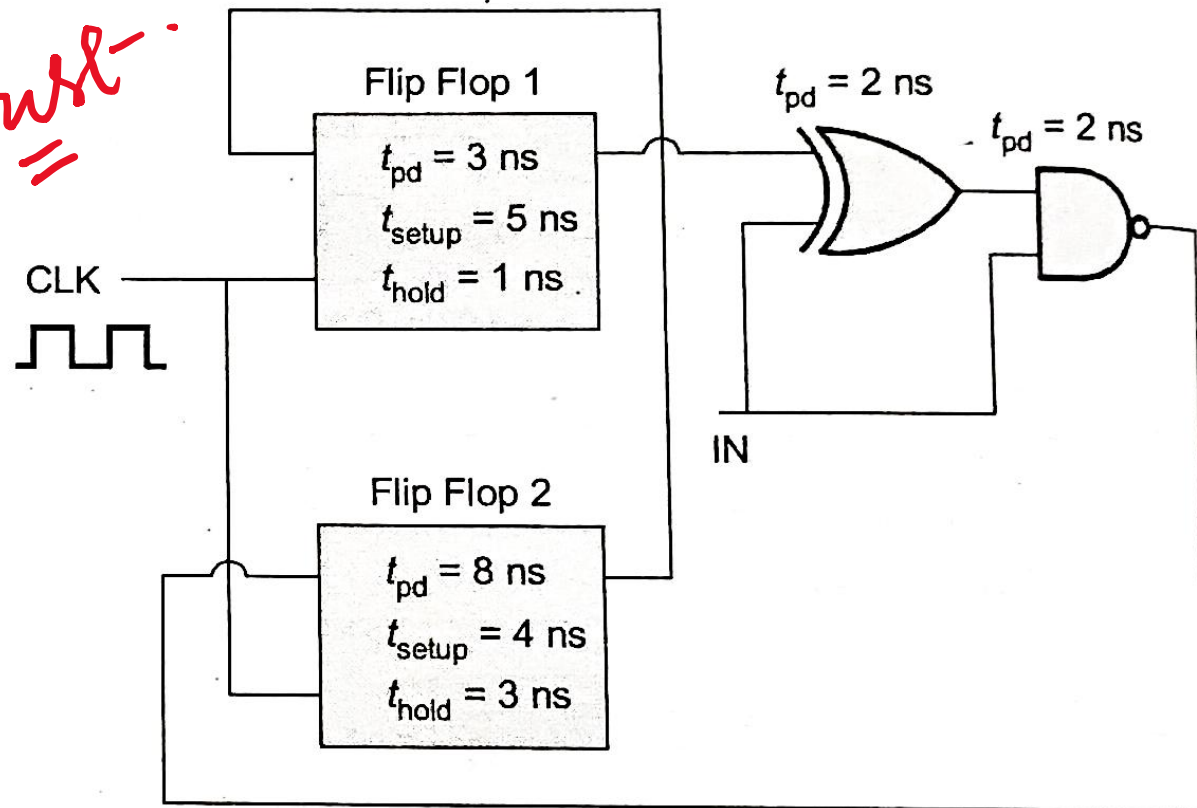
[2019 : 1 Mark]

→ Modulus of this counter = $N = 2^n = 2^3 - 1$

∴ frequency = $\frac{12 \text{ kHz}}{3} = \boxed{4 \text{ kHz}}$ $= 4 - 1 = 3$

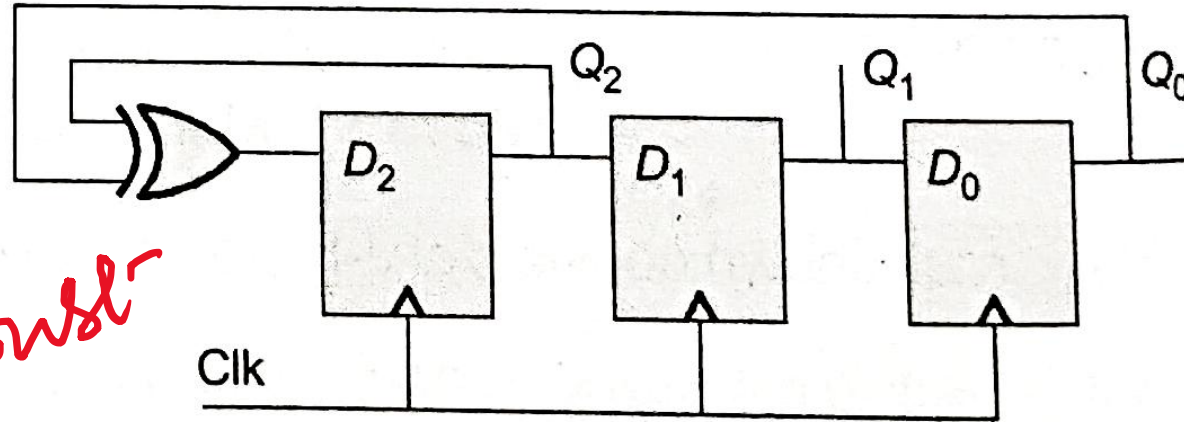
Frequency
Divides
ckt.
F.D.

5.53 For the components in the sequential circuit shown below, t_{pd} is the propagation delay, t_{setup} is the setup time, and t_{hold} is the hold time. The maximum clock frequency (rounded off to the nearest integer), at which the given circuit can operate reliably, is ____ MHz.



[2020 : 2 Marks]

5.54 The propagation delay of the exclusive-OR (XOR) gate in the circuit in the figure is 3 ns. The propagation delay of all the flip-flops is assumed to be zero. The clock (Clk) frequency provided to the circuit is 500 MHz.



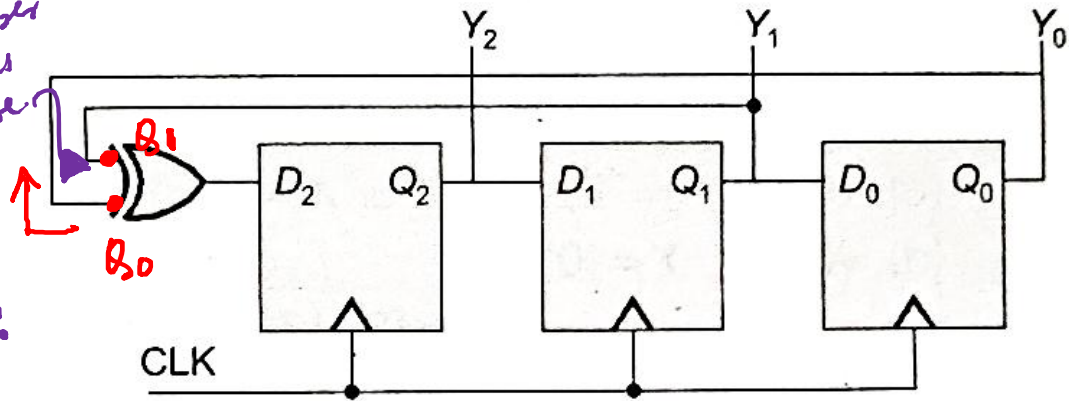
answer

Starting from the initial value of the flip-flop outputs $Q_2 Q_1 Q_0 = 111$ with $D_2 = 1$, the minimum number of triggering clock edges after which the flip-flop outputs $Q_2 Q_1 Q_0$ becomes 100 (in integer) is _____.

[2021 : 2 Marks]

5.46 A three bit pseudo random number generator is shown. Initially the value of output $Y = Y_2 Y_1 Y_0$ is set to 111. The value of output Y after three clock cycles is

Remember this is XOR gate NOT and gate !!



(a) 000

(b) 001

(c) 010

(d) 100

[2015 : 2 Marks, Set-3]

MSB. $Y = Y_2 Y_1 Y_0$ LSB.
1 1 1

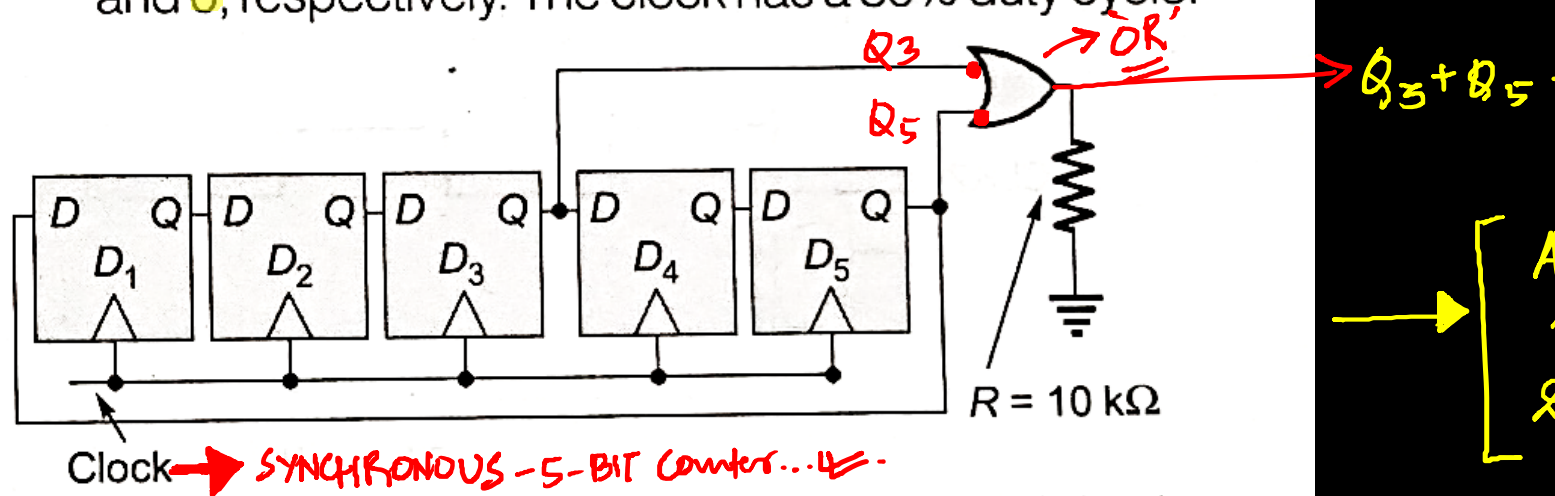
@ 1st CP $\rightarrow$ 1 1 0

@ 2nd CP $\rightarrow$ 1 0 1

@ 3rd CP $\rightarrow$ 1 0 0

3-BIT
SYNCHRONOUS
down counter.
=

5.47 Assume that all the digital gates in the circuit shown in the figure are ideal, the resistor $R = 10\text{ k}\Omega$ and the supply voltage is 5 V . The D flip-flops D_1 , D_2 , D_3 , D_4 and D_5 are initialized with logic values $0, 1, 0, 1$ and 0 , respectively. The clock has a 30% duty cycle.

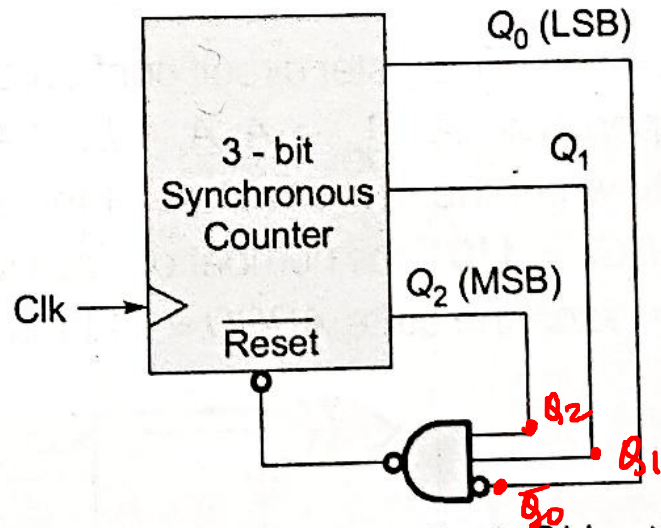


The average power dissipated (in mW) in the resistor R is 1.5 mW [2016 : 1 Mark, Set-2]

$$\begin{aligned} \text{Average power Dissipated} &= P = \frac{V^2}{R} \times \frac{T_{\text{ON}}}{\text{Total time}} \\ &= \frac{5^2}{10\text{ k}\Omega} \times \frac{30\%}{100\%} = \boxed{1.5\text{ mW}} \end{aligned}$$

ANS:

- 5.48 For the circuit shown in the figure, the delay of the bubbled NAND gate is 2 ns and that of the counter is assumed to be zero.



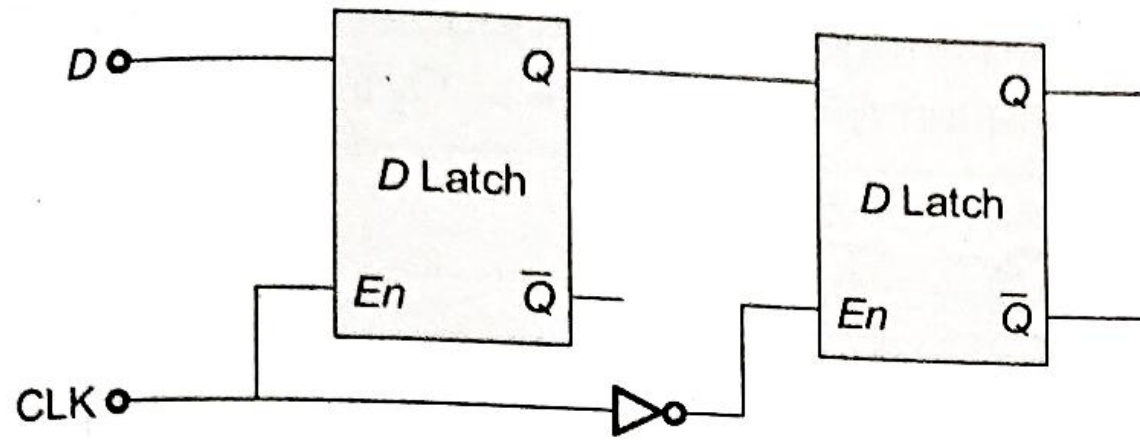
If the clock (Clk) frequency is 1 GHz, then the counter behaves as a

- (a) mod-5 counter (b) mod-6 counter
(c) mod-7 counter (d) mod-8 counter

[2016 : 2 Marks, Set-3]

	2	1	
Q_0	Q_1	Q_2	
0	0	0	-
0	0	1	-
0	1	0	-
-	0	1	-
-	1	0	0
-	1	0	1
-	1	1	0
-	1	1	1

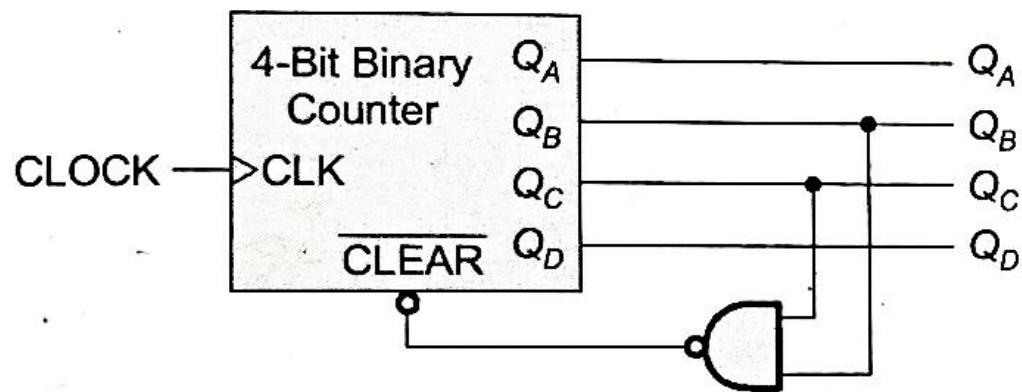
5.42 The circuit shown in the figure is a



- (a) Toggle Flip Flop
- (b) JK Flip Flop
- (c) SR Latch
- (d) Master-Slave D Flip Flop

[2014 : 1 Mark, Set-3]

5.43 A mod- n counter using a synchronous binary up-counter with synchronous clear input is shown in the figure. The value of n is 7. (Ans)



[2015 : 1 Mark, Set-2]

4 variables. Means max count would be $2^4 - 1 = 16 - 1 = 15$

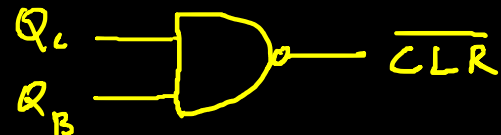
0000 ✓

1111 ✓

LSB → 0 1 1 0 ← MSB

[starting bit]

$Q_A \quad Q_B \quad Q_C \quad Q_D$
0 1 1 0



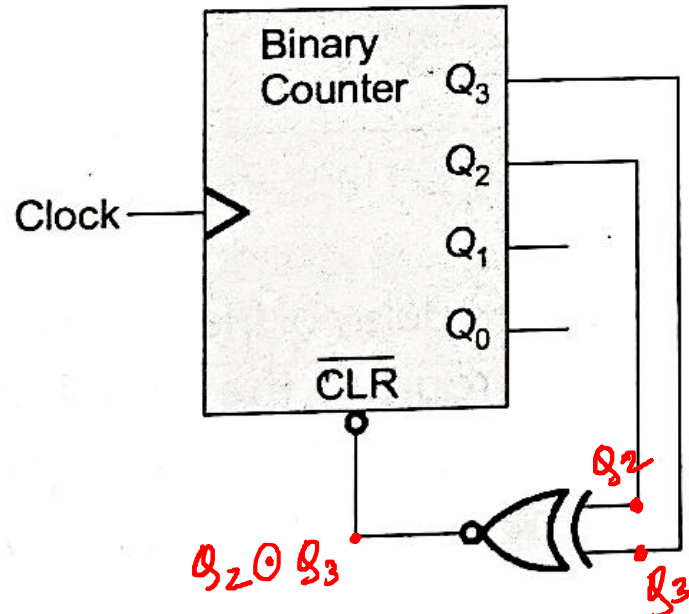
* If $Q_B = Q_C = 1$; then only 0% of NAND gate can be '0' that'll in turn ON the CLR pin as it's ACTIVE LOW i.e. ...

Q_A	Q_B	Q_C	Q_D
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

0 → ①
1 → ②
2 → ③
3 → ④
4 → ⑤
5 → ⑥
6 → ⑦

0-6
7-11

5.44 The figure shows a binary counter with synchronous clear input. With the decoding logic shown, the counter works as a



(a) mod-2 counter

✓ (c) mod-5 counter

(b) mod-4 counter

(d) mod-6 counter

[2015 : 2 Marks, Set-2]

Now, Coming to the XNOR gate
Truth Table; We get...

Q_2	Q_3	$Q_2 \odot Q_3$
0	0	1
0	1	0
1	0	0
1	1	1

0 →
1 →
2 →
3 →
4 →

0-4 =

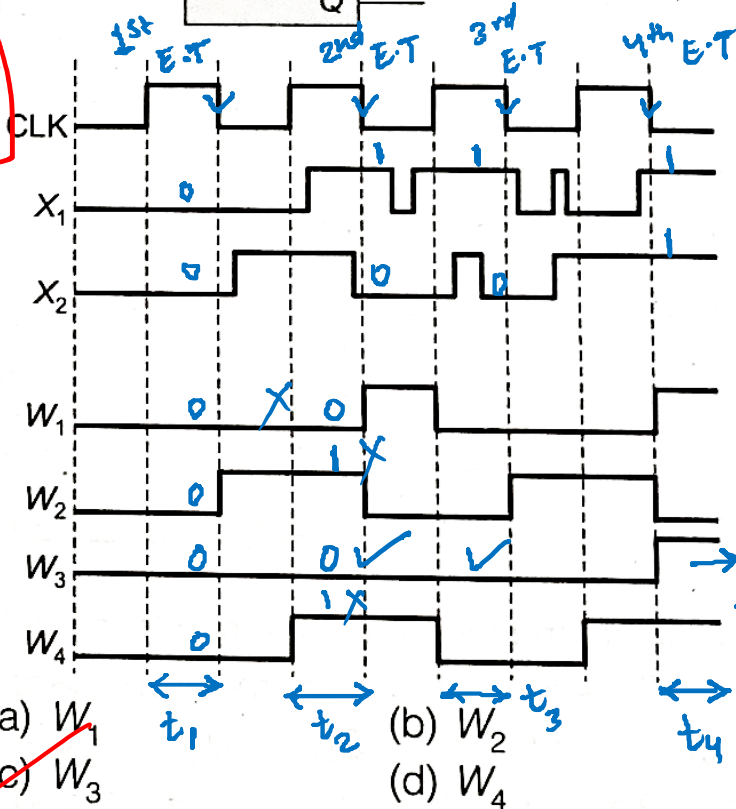
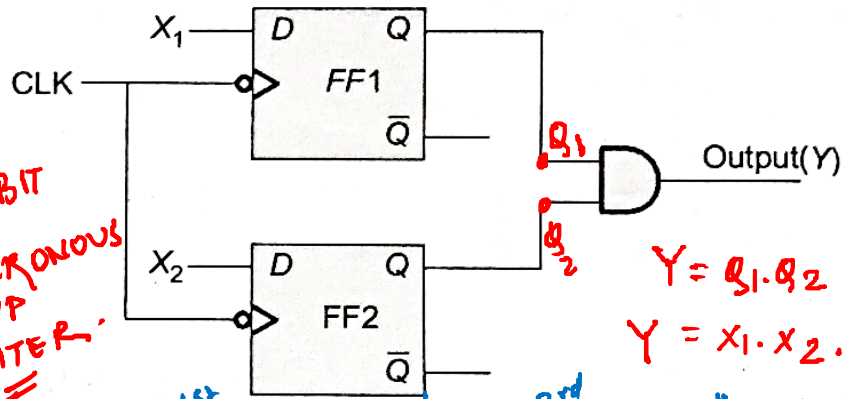
5

MOD-5
Counter !!

Q_0	Q_1	Q_2	Q_3
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

5.40 In the circuit shown, choose the correct timing diagram of the output (Y) from the given waveforms W_1 , W_2 , W_3 and W_4 .

2-BIT
SYNCHRONOUS
UP
COUNTER.
[-ve slope
Triggered]
CLK.

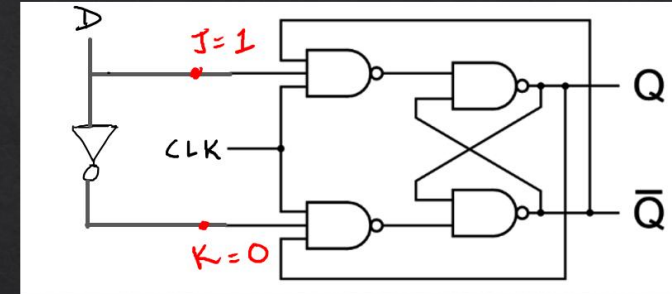


[2014 : 2 Marks, Set-2]

D FlipFlop

Input		Output	
J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

CHARACTERISTIC TABLE



$D =$

J	K	Q_n
0	0	Q_n
0	1	0
1	0	1
1	1	1

TRUTH-TABLE

$$Y_1 = X_1 \cdot X_2 = 0 \cdot 0 = 0 \quad [@ t_1]$$

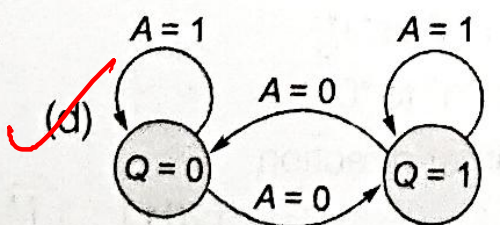
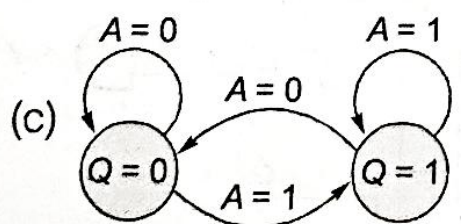
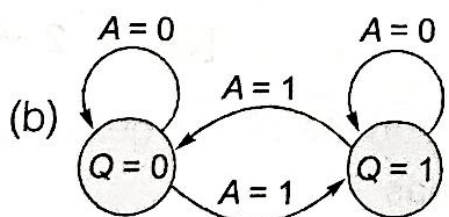
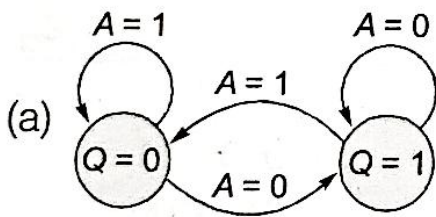
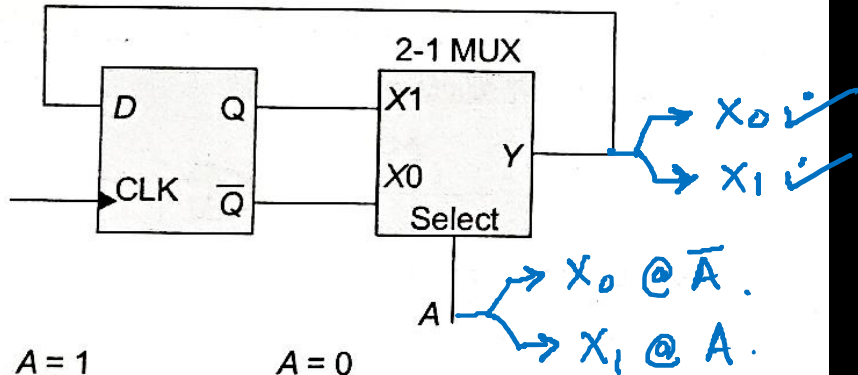
$$Y_2 = X_1 \cdot X_2 = 1 \cdot 0 = 0 \quad [@ t_2]$$

$$Y_3 = X_1 \cdot X_2 = 1 \cdot 0 = 0 \quad [@ t_3]$$

$$Y_4 = X_1 \cdot X_2 = 1 \cdot 1 = 1 \quad [@ t_4] \rightarrow \underline{W_3} \text{ (ANS)}$$

$\rightarrow \text{ANS.}$

5.38 The state transition diagram for the logic circuit shown in

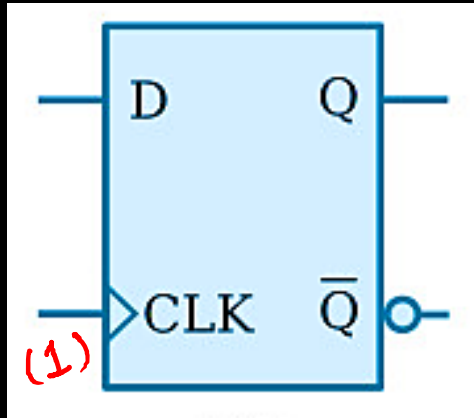


[2012 : 2 Marks]

when $A = 0$;
 $Y = X_0$ and $\bar{Q} = D$.

Similarly ;
= when $A = 1$;
 $Y = X_1$ and $Q = D$.

WAVEFORMS ANALYSIS



+ve Edge Triggered

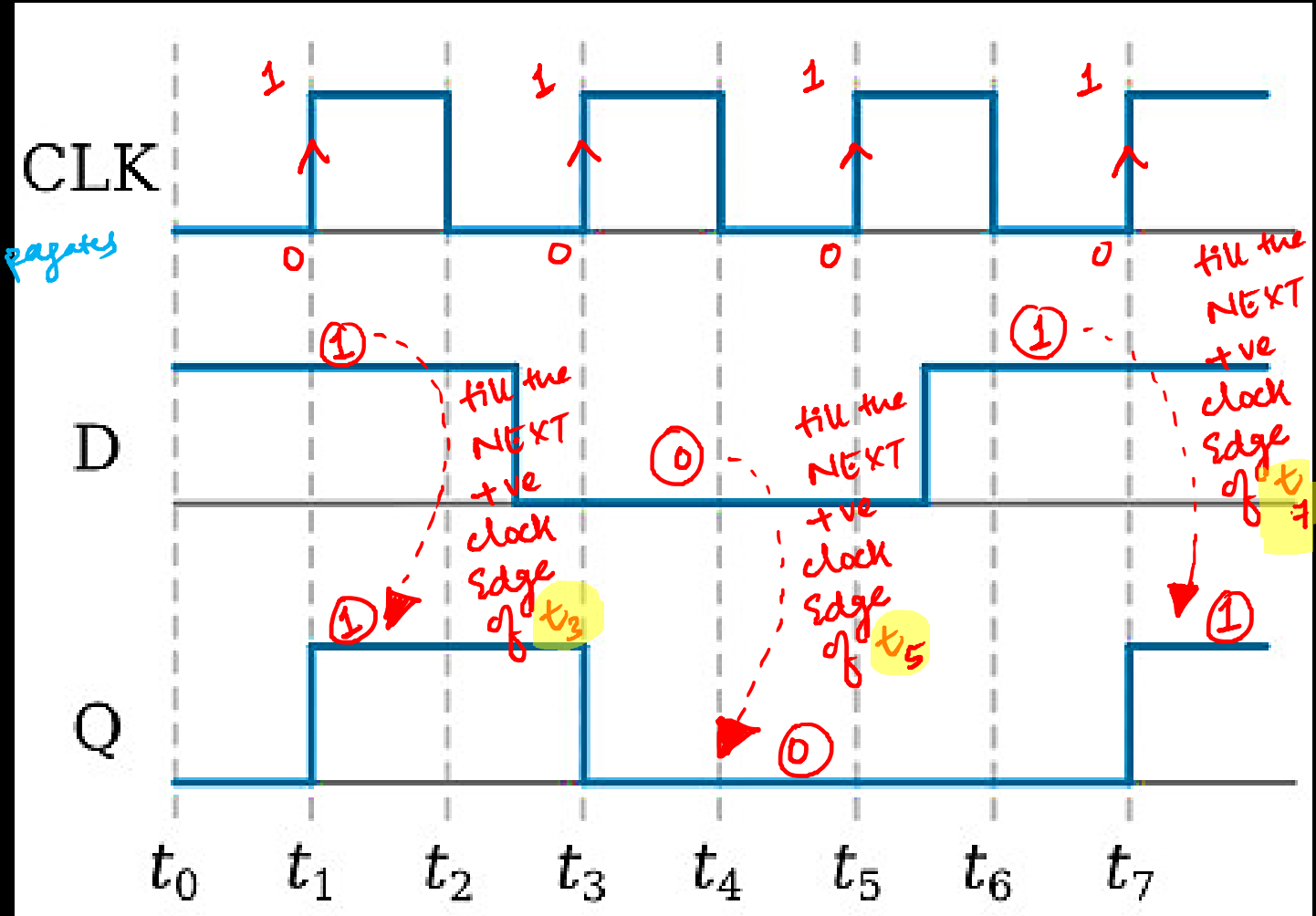
CONCEPT
Because 'D' FF acts as a BUFFER; it just copies/propagates the values.
① → ①
① → ①
① → ①
① → ①

	CLK	D	Q _{next}	Comment
↑	Rising edge	0	0	Store 0
↑	Rising edge	1	1	Store 1
×	Non-rising	X	Q	No change

Q_{next} - "after the clock transition" output

Q - the current output

X - the signal is irrelevant

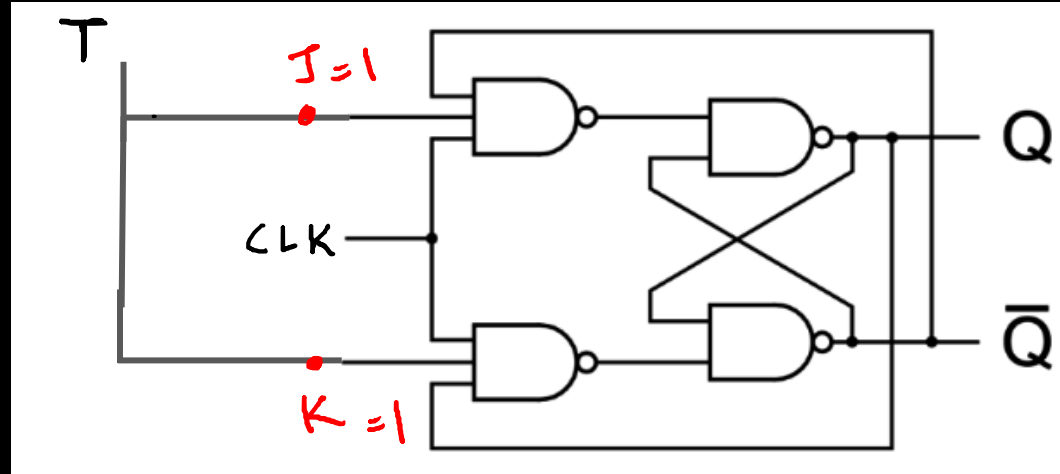


T FlipFlop

Input		Output	
J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

$\left. \begin{matrix} 0 \\ 0 \\ 0 \end{matrix} \right\} Q_n$
 $\left. \begin{matrix} 0 \\ 0 \end{matrix} \right\} 0$
 $\left. \begin{matrix} 1 \\ 1 \end{matrix} \right\} 1$
 $\left. \begin{matrix} 0 \rightarrow 1 \\ 1 \rightarrow 0 \end{matrix} \right\} \bar{Q}_n$

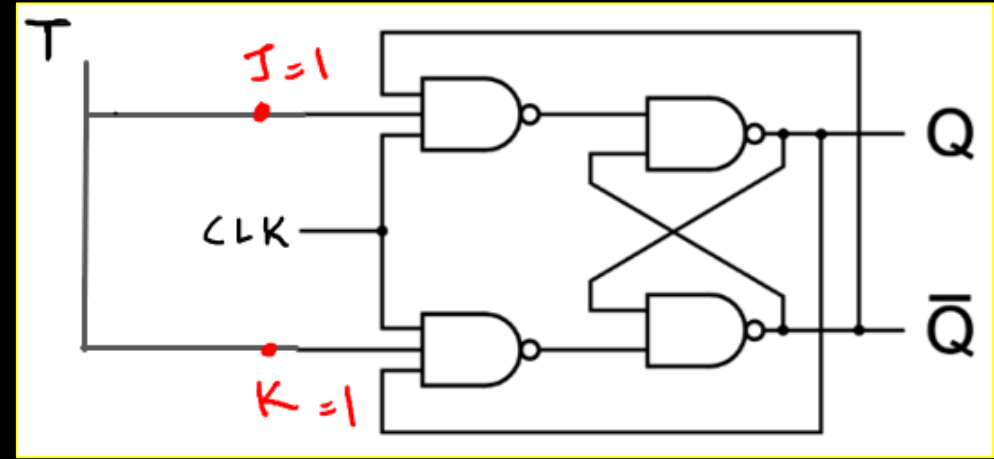
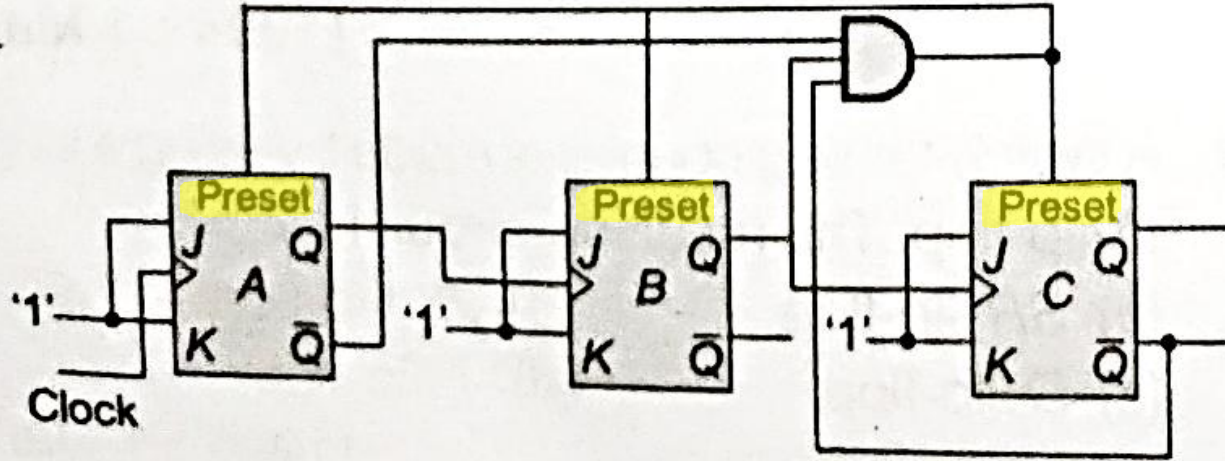
CHARACTERISTIC TABLE



J	K	Q_n
0	0	Q_n
0	1	0
1	0	1
1	1	$\bar{Q}_n$

TRUTH-TABLE

5.14 The ripple counter shown in the figure works as a



- (a) mod-3 up counter
- (b) mod-5 up counter
- (c) mod-3 down counter
- ☒ (d) mod-5 down counter

- According to the Question;

✓ PRESET is an ACTIVE HIGH pin.

Now; "AND" gate y_p would be 1; so

as to SET the value of the FF's to 111.

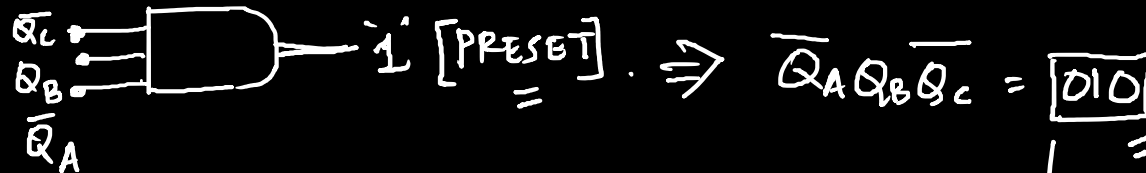
what Means; it's a DOWN COUNTER starting from (11).

$$\begin{array}{r} 111 \text{ --- } 1 \\ 110 \text{ --- } 2 \\ 101 \text{ --- } 3 \\ 100 \text{ --- } 4 \\ 011 \text{ --- } 5 \\ \hline 010 \\ 001 \end{array}$$

MOD-5
DOWN
COUNTER

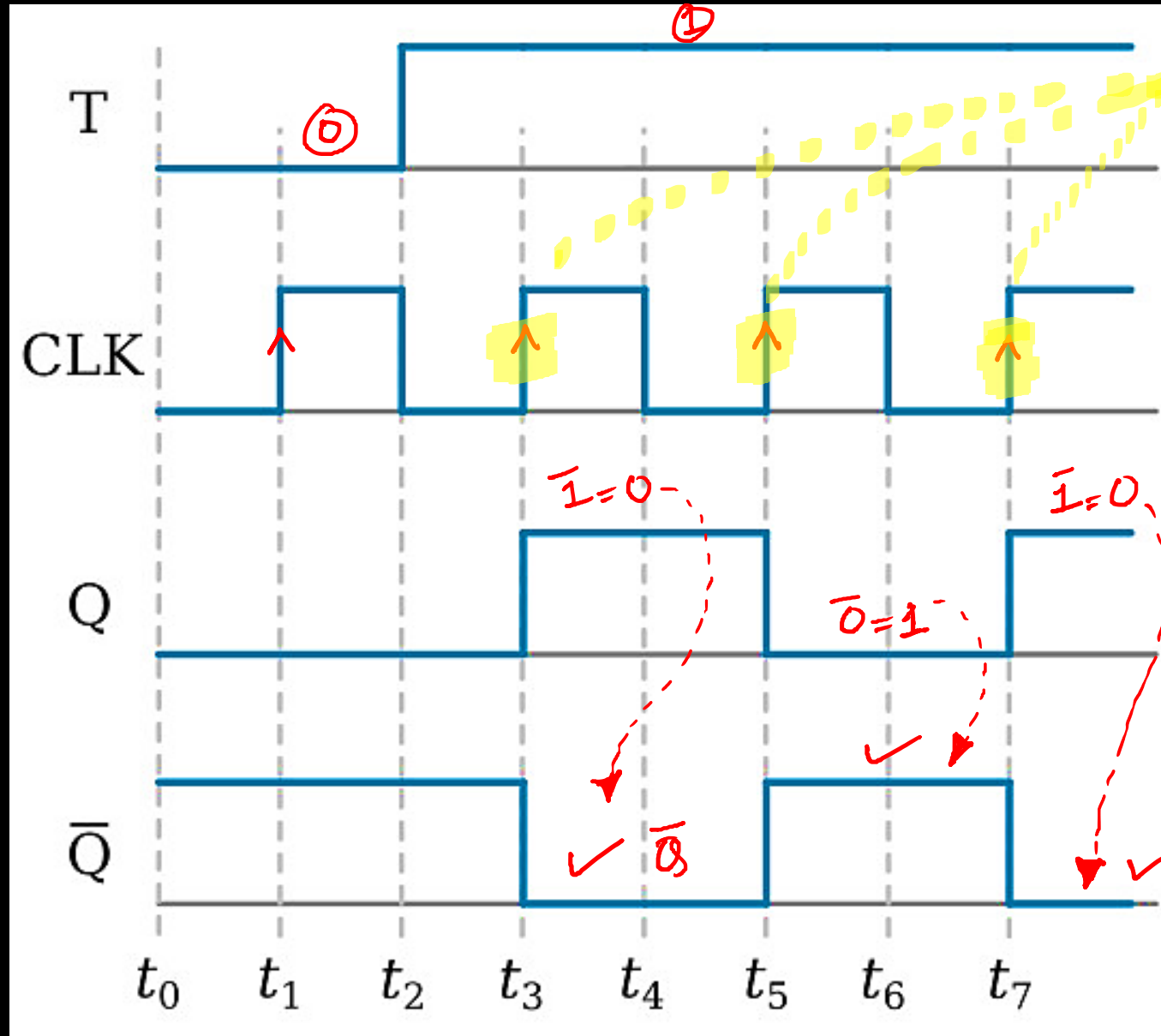
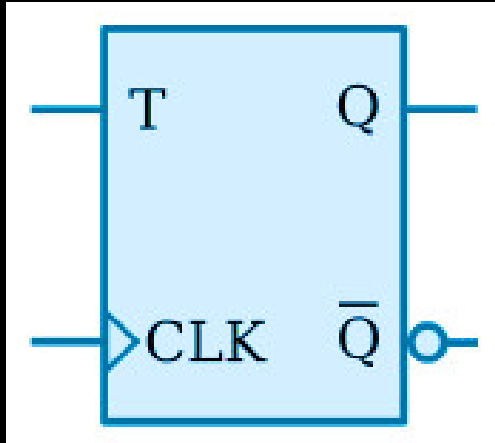
<u>J</u>	<u>K</u>	<u>Qn</u>
0	0	Qn
0	1	0
1	0	1
1	1	$\bar{Q}_n$

↳ i.e. the UNDESIRABLE state...



↳ i.e. the UNDESIRABLE state...

WAVEFORMS ANALYSIS



CONCEPT:-

• At each +ve Edge of clock... whatever be the value either '0' or '1'; it will Toggle.

- (i) $0 \rightarrow 1$
- (ii) $1 \rightarrow 0$

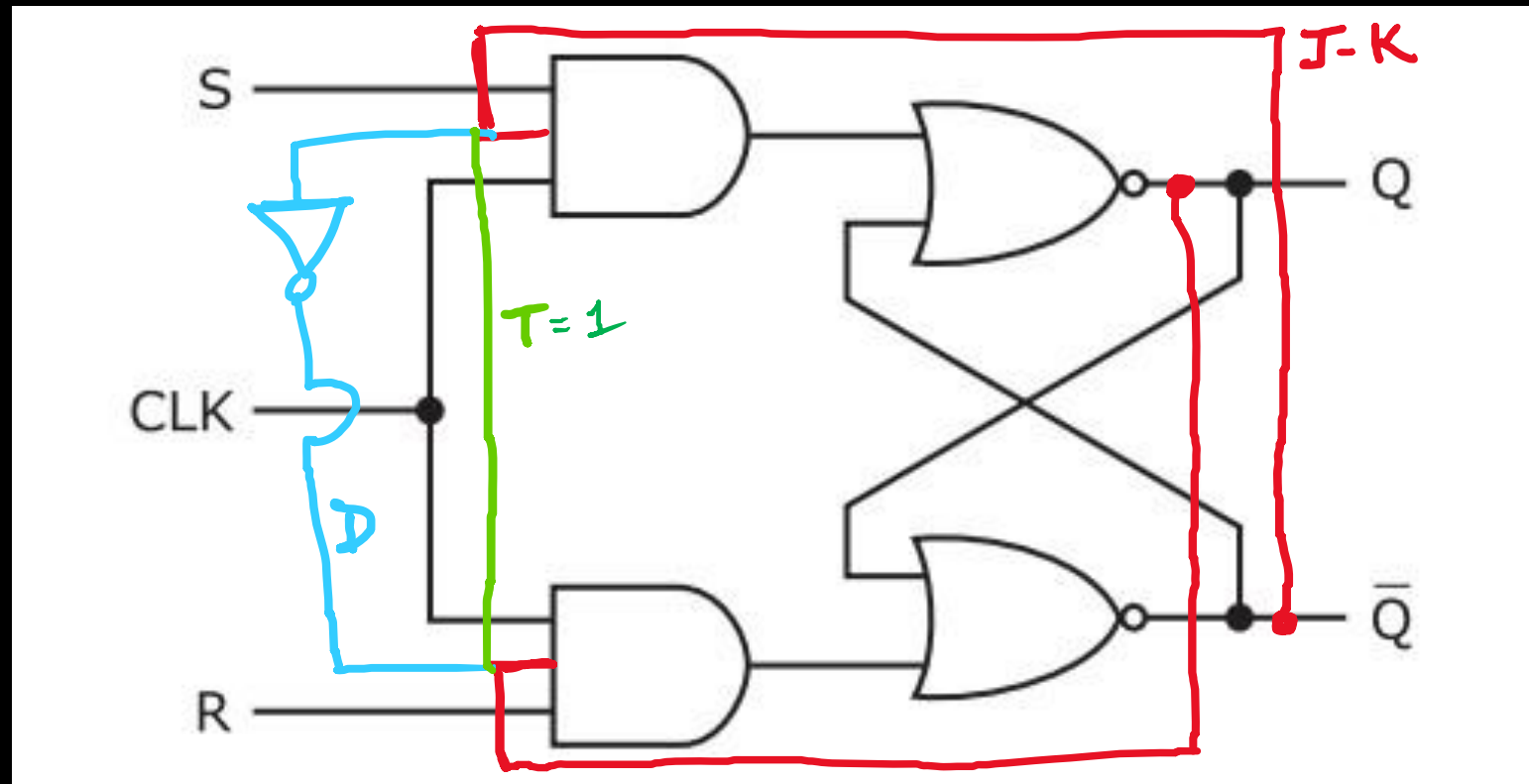
Truth table

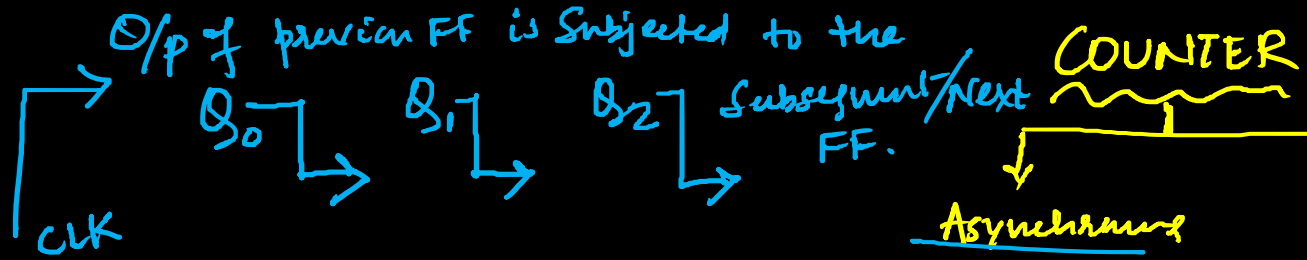
CLK	T	Q_{next}	Comment
Rising edge	0	Q	Hold state
Falling edge	0	Q	Hold state
Rising edge	1	$\bar{Q}$	Toggle
Falling edge	1	Q	No change

Q_{next} - "after the clock transition" output

Q - the current output

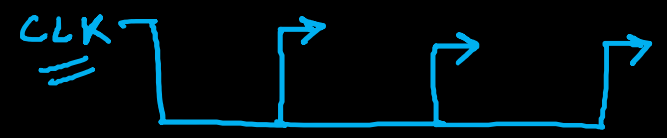
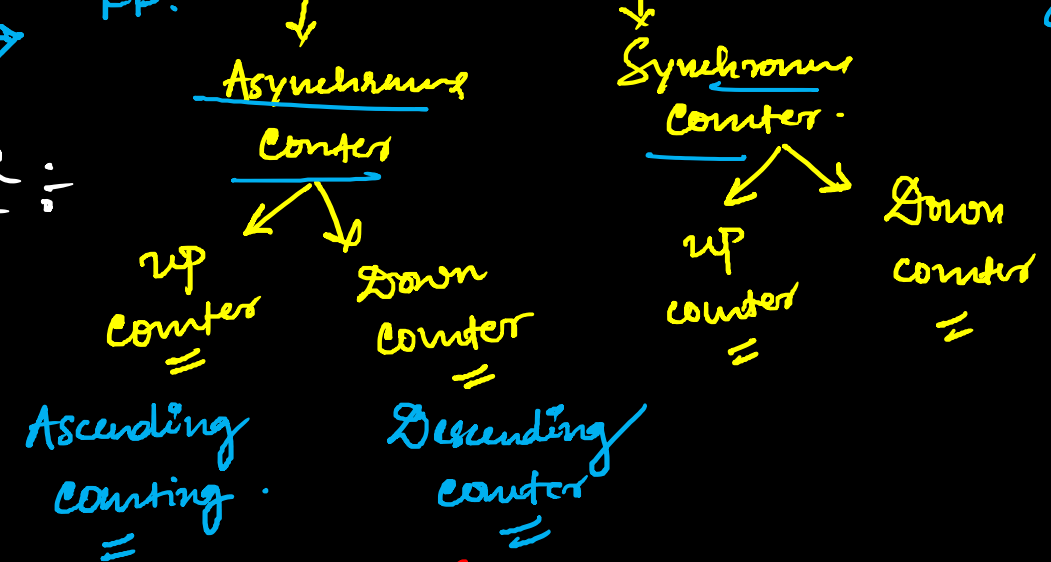
... Summary ...





MODULUS OF A COUNTER :-

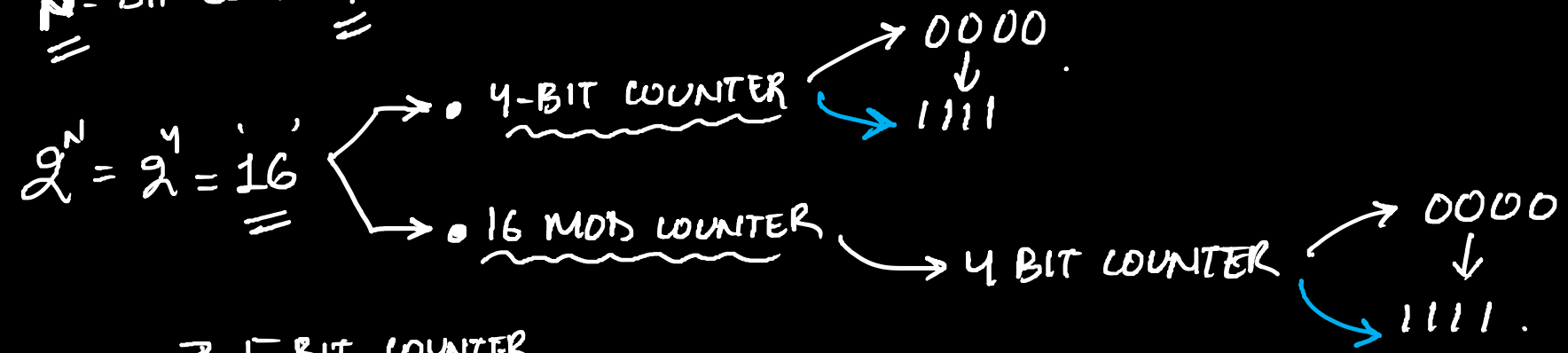
No. of states passed by the counter before going/reaching its Initial state.



here, "CLK" is given Individually to Each and Every FlipFlop.

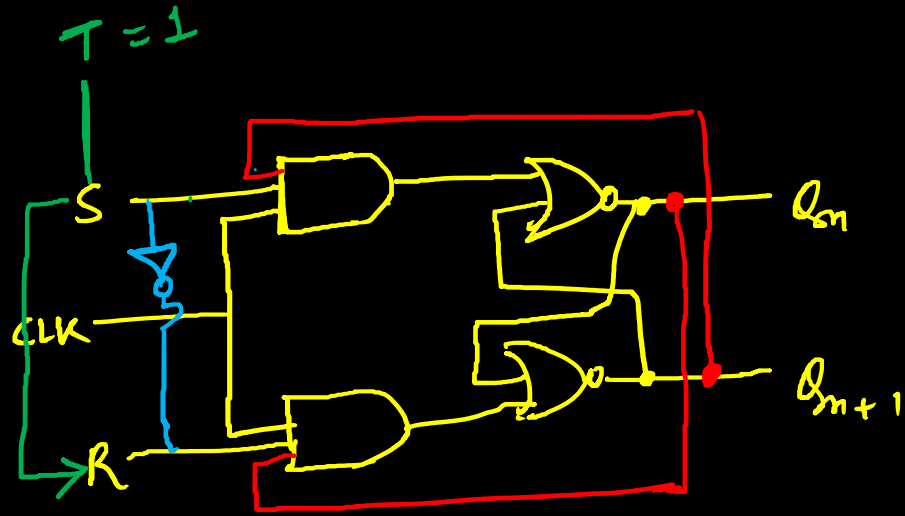
eg:- 4-BIT COUNTER
N-BIT COUNTER

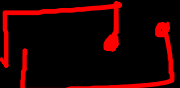
$0 \rightarrow 2^n - 1$ $2^n - 1 \rightarrow 0$



$N=5 \rightarrow$ 5 BIT COUNTER
 2^5 MOD COUNTER / 32 MOD COUNTER

If we've 2 counters of 'm' MOD & 'n' MOD; then the equivalent MODULUS of the aggregate counter = 'm x n'.



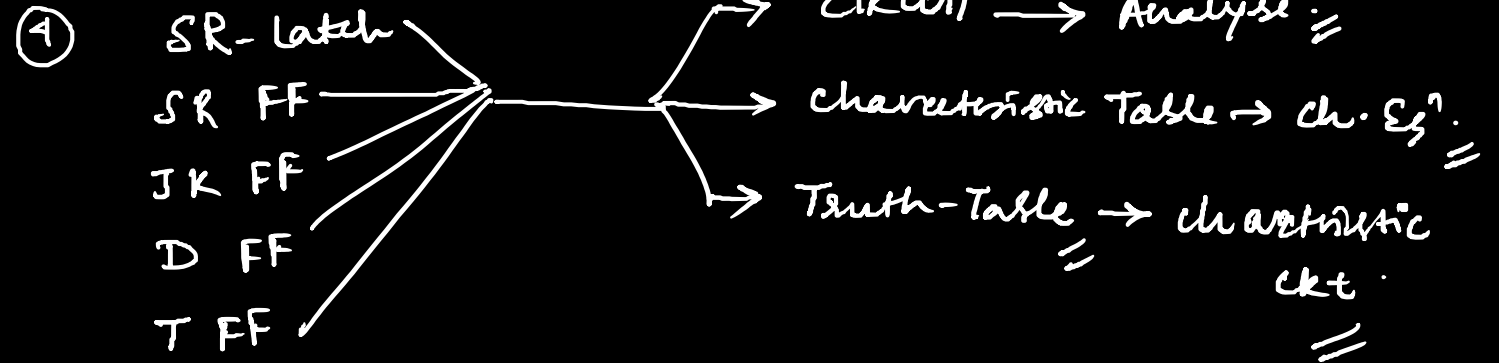
0. S-R flipflop. ✓ + latch
1.  $\equiv$ D flipflop. ✓
2.  $\equiv$ J-K flipflop ✓
3. $S=R=1=\textcircled{T}$ $\equiv$ T flipflop. ✓

REVISION

- ① CLK $\begin{cases} \rightarrow \text{Edge triggering} \\ \rightarrow \text{Level triggering} \end{cases}$

② Synchronous / Asynchronous $\rightarrow$ TRADE OFFS.

③ Latches Vs FlipFlops.



- ④ SR-latch
- SR FF
- JK FF
- D FF
- T FF

Summary

⑤ We did some J-K ff Wale Gate Question $\rightarrow$ FD $\rightarrow$ CS [dent] $\rightarrow$ Counters.

SEQUENTIAL
chts

Completed $\nabla \nabla$
 $\equiv$ 0 0